```
pip install torch transformers datasets peft accelerate
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: peft in /usr/local/lib/python3.12/dist-packages (0.18.1)
Requirement already satisfied: accelerate in /usr/local/lib/python3.12/dist-packages (1.13.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
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Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
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Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.7
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Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.2
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Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->data
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
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Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
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Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datase
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hugg
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Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0
```

```
import torch
import pandas as pd
from datasets import Dataset
from transformers import AutoTokenizer, AutoModelForSequenceClassification
from transformers import TrainingArguments, Trainer
from peft import LoraConfig, get_peft_model
```

```
model_name = "distilbert-base-uncased"

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForSequenceClassification.from_pretrained(
    model_name,
    num_labels=2
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
```

```
config.json: 100%  483/483 [00:00<00:00, 5.01kB/s]

tokenizer_config.json: 100%  48.0/48.0 [00:00<00:00, 604B/s]

vocab.txt: 232k/? [00:00<00:00, 1.77MB/s]

tokenizer.json: 466k/? [00:00<00:00, 3.62MB/s]

model.safetensors: 100%  268M/268M [00:02<00:00, 185MB/s]

Loading weights: 100%  100/100 [00:00<00:00, 233.90it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
```

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                      | Status     |
-------------------------+------------+-
vocab_layer_norm.weight  | UNEXPECTED |
vocab_transform.weight   | UNEXPECTED |
vocab_layer_norm.bias    | UNEXPECTED |
vocab_projector.bias     | UNEXPECTED |
vocab_transform.bias     | UNEXPECTED |
classifier.weight        | MISSING    |
pre_classifier.bias      | MISSING    |
pre_classifier.weight    | MISSING    |
classifier.bias          | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
```

```
data = pd.read_csv("sentiment_dataset_peft_lab19.csv")

dataset = Dataset.from_pandas(data)

def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)

tokenized_dataset = dataset.map(tokenize_function)

dataset = tokenized_dataset.train_test_split(test_size=0.2)

train_dataset = dataset["train"]
test_dataset = dataset["test"]
```

```
Map: 100%  300/300 [00:01<00:00, 507.84 examples/s]
```

```
lora_config = LoraConfig(
    r=8,
    lora_alpha=16,
    target_modules=["q_lin", "v_lin"],
    lora_dropout=0.1,
    bias="none",
    task_type="SEQ_CLS"
)
```

```
peft_model = get_peft_model(model, lora_config)

peft_model.print_trainable_parameters()
```

```
WARNING:torchao.kernel.intmm:Warning: Detected no triton, on systems without Triton certain kernels will not work
trainable params: 739,586 || all params: 67,694,596 || trainable%: 1.0925
```

```
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-4,
    per_device_train_batch_size=8,
    num_train_epochs=1,
    logging_steps=10,
    save_strategy="no"
```

```
)

trainer = Trainer(
    model=peft_model,
    args=training_args,
    train_dataset=train_dataset
)

trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true bu
  super().__init__(loader)
```

[26/30 07:52 < 01:18, 0.05 it/s, Epoch 0.83/1]

Step	Training Loss
10	0.688281
20	0.575099

[30/30 09:34, Epoch 1/1]

Step	Training Loss
10	0.688281
20	0.575099
30	0.509483

```
TrainOutput(global_step=30, training_loss=0.5909542719523112, metrics={'train_runtime': 599.28, 'train_samples_per_second':
0.4, 'train_steps_per_second': 0.05, 'total_flos': 32337457643520.0, 'train_loss': 0.5909542719523112, 'epoch': 1.0})
```

```
trainer.evaluate(eval_dataset=test_dataset)

text = "The movie was fantastic and inspiring!"

inputs = tokenizer(text, return_tensors="pt")

outputs = peft_model(**inputs)

prediction = torch.argmax(outputs.logits, dim=1)

print("Predicted label:", prediction.item())
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true bu
  super().__init__(loader)
```

[8/8 00:39]

```
Predicted label: 1
```

Start coding or generate with AI.